

2024

JetBrains survey of 23,262 developers
worldwide

State of Developer Ecosystem 2024: Language Trends, Satisfaction, and Strategic Shifts

A data-driven view of how language usage is changing, what developers actually prefer, and where leaders should adapt technology strategy and hiring priorities.

23,262

developers surveyed
worldwide

20

languages tracked

8 yrs

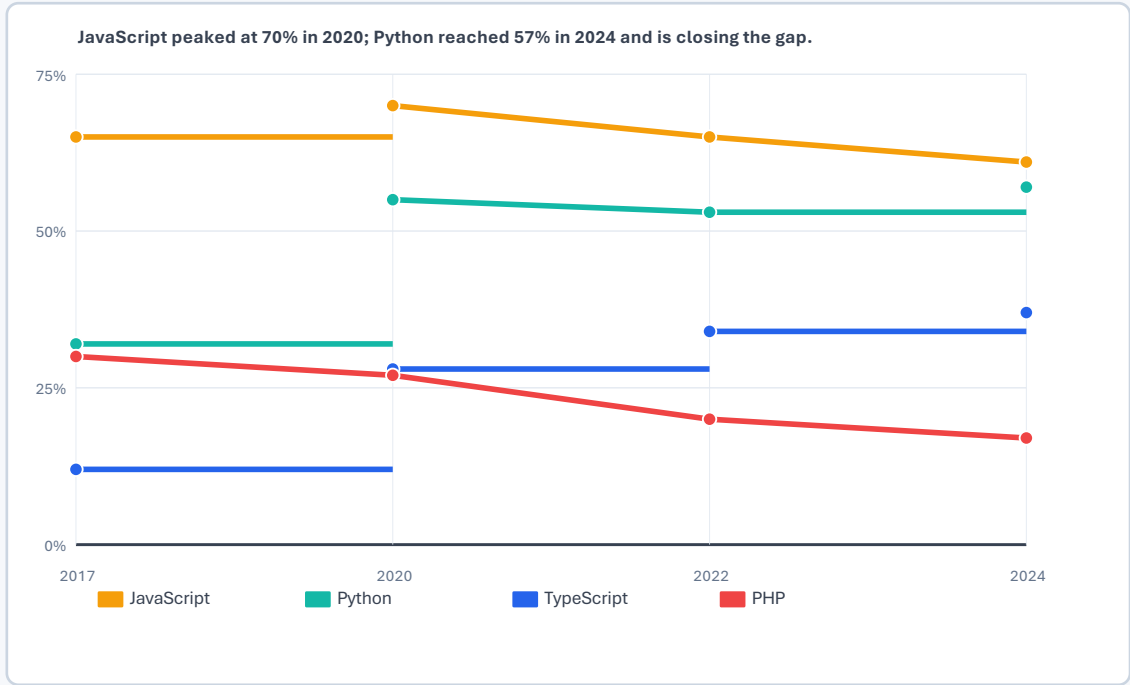
usage trend history



Narrative arc: usage shifts → developer preference → strategic actions

Python Closes the Gap: 8 Years of Language Usage Shifts

Usage data from JetBrains' annual developer ecosystem survey; shares are developers using each language.



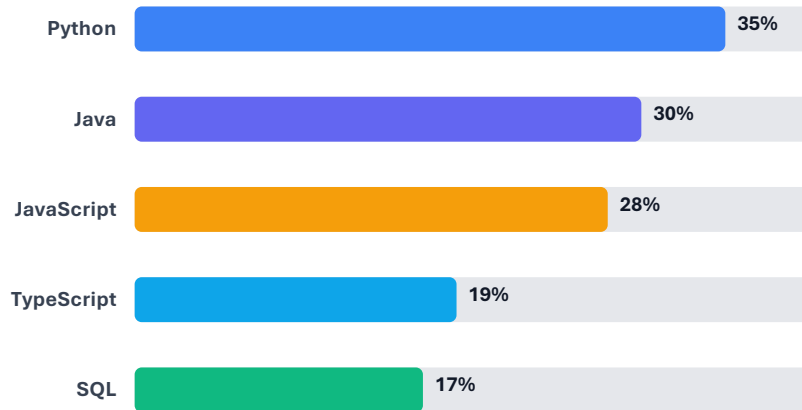
Top 8 languages: usage comparison				
Language	2017	2020	2022	2024
JavaScript	65%	70%	65%	61%
Python	32%	55%	53%	57%
HTML/CSS	60%	61%	54%	51%
SQL	42%	56%	49%	48%
Java	47%	54%	48%	46%
TypeScript	12%	28%	34%	37%
Go	8%	19%	19%	18%
Rust	--	7%	9%	11%

Reshuffle signal: Python and TypeScript are up; PHP declines sharply while JavaScript remains most used overall.

Python Is Now the #1 Primary Language at 35%

Primary language choices show where ecosystem commitment is strongest; adoption intent remains modest.

Primary language breakdown

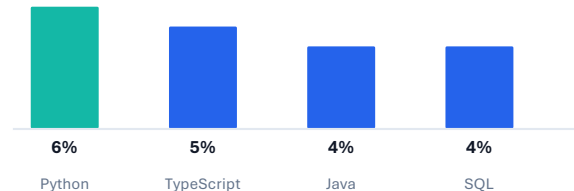


Python leads primary-language share, ahead of Java and JavaScript.

Comfort zone effect

Developers show strong loyalty to their chosen ecosystems. Adoption intent remains below 6% across all languages.

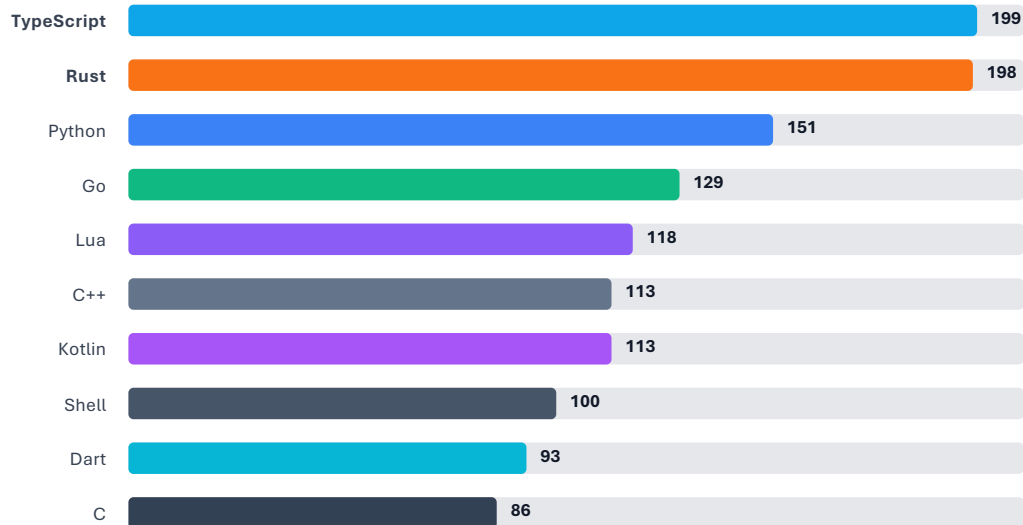
Planned adoption is modest



TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198

Satisfaction scores from the JetBrains survey show modern tooling and safety commanding developer enthusiasm.

Top 10 language satisfaction scores



199

TypeScript
leads satisfaction

Leadership readout

Type-safe, modern-tooling languages dominate the top spots — an indicator for retention and platform investment priorities.

Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Language choice follows application domain more than generic preference.

Domain leaders heatmap

Domain	Leader	Share
AI / Machine Learning	Python	31%
Data Science / Analytics	Python	28%
Data Processing & Analytics	Python	17%
User Interface	TypeScript	62%
User Interface	JavaScript	58%
APIs & Services	Java	49%
Application Logic & Workflows	Java	55%
System Tools & Components	C++	27%
Hardware Interfacing	C++	21%
Rendering & Graphics	C++	15%
Integrating with APIs & Services	Python	38%

Portfolio implication

Python anchors AI and data.
TypeScript / JavaScript anchor
UI. Java anchors API and
workflow platforms. C++
remains critical for systems,
hardware, and graphics.

Strategy should map hiring
and tooling to domain
specialization rather than
treating “language popularity”
as one ranking.

What This Means for Your Technology Strategy in 2025

Actionable priorities for engineering leaders, CTOs, and developer experience teams.

From popularity to portfolio design

The language landscape is not converging on one winner. It is specializing: pick languages for domain fit, talent availability, and developer experience.

Operating model

Segment platforms by domain, measure developer satisfaction, and keep hiring plans aligned with language momentum.

1

Invest in Python for AI/ML teams

Python leads AI / machine learning at 31%, data science at 28%, and data processing at 17%.

2

Standardize TypeScript for UI and full-stack

TypeScript leads UI at 62% and tops developer satisfaction with a score of 199.

3

Treat Rust and Go as infrastructure growth plays

Go stabilized around 18% usage after rapid growth; Rust continues climbing to 11%.

4

Plan carefully around shrinking legacy pools

PHP dropped from 30% in 2017 to 17% in 2024; Ruby fell from 10% to 4%.

5

Use satisfaction as a retention signal

Type-safe, modern-tooling languages dominate the top satisfaction spots.